Task 2 :

Report: Category Flow Mapping Agent Using Gemini API

Task 2 involved building a Python-based AI agent that classifies incident descriptions into predefined attack categories. The agent utilizes Google's Gemini models (version 1.5 and 2.0) for inference and includes a robust evaluation framework comparing model outputs against manually labeled ground truth using selected metrics.

Functionality

- **Input**: A textual incident description.
- **Prompt**: Contains formal definitions of "Attack" and all 8 category codes:
    - AAT: Advanced & Emerging Tech Attacks
    - BPA: Biological & Pandemic Attacks
    - DCA: Digital & Cyber Attacks
    - EFA: Economic & Financial Attacks
    - NDA: Natural Disaster Attacks
    - OIA: Operational & Industrial Attacks
    - PSI: Physical Security & Infrastructure Attacks
    - SPI: Socio-Political & Influence Attacks
- **Output**: All applicable category codes strictly in JSON list format.
- **Model Selection**: Users can choose between gemini-1.5-flash and gemini-2.0-flash per run.

Evaluation Metrics

To evaluate how closely the predicted category flow matches the ground truth, the following metrics were used:

a. F1 Score (Micro & Macro)

- **Purpose**: Measures the balance between precision and recall.
    - **Micro-F1** aggregates contributions of all classes to compute an overall score (better for frequent categories).
    - **Macro-F1** calculates F1 independently for each label and averages them (better for class balance).
- **Relevance**: Captures both overprediction and underprediction in a multi-label scenario.

b. Jaccard Similarity Score (Micro & Macro)

- **Purpose**: Measures set overlap between predicted and actual category sets.
    - **Micro-Jaccard** reflects overall prediction accuracy across all labels.

    - **Macro-Jaccard** gives equal weight to each sample's prediction quality.
- **Relevance**: Emphasizes the importance of predicting the exact set of categories correctly.

Evaluation Results

Metric	Gemini 2.0	Gemini 1.5
Micro F1 Score	**0.7431**	**0.7143**
Macro F1 Score	**0.6274**	**0.5703**
Micro Jaccard Score	**0.5912**	**0.5556**
Macro Jaccard Score	**0.4909**	**0.4433**

Output Analysis

Performance Comparison

- **Gemini 2.0 outperforms 1.5** across all metrics, showing better multi-label classification performance.
- **Micro-level scores** indicate improved handling of common categories.
- **Macro-level improvements** highlight Gemini 2.0's ability to better address less frequent categories as well.

Observations

- Gemini 2.0 is more effective at **capturing multi-category incidents** and **understanding downstream implications**, such as how a physical or natural attack may affect digital systems.
- Gemini 1.5 tends to predict only the **primary or most obvious category**, missing secondary effects.

Limitations of Metrics I used in the process

Limitation	Description
Order Ignorance	F1 and Jaccard ignore the temporal or causal order of predicted categories.
Partial Matching	Metrics treat partial matches the same, regardless of importance (e.g., missing "SPI" may be more impactful than missing "OIA").
Equal Weighting in Macro Scores	Assumes all categories are equally important, which may not align with real-world impact.